복사전달 특강
Special Topics in Radiative Transfer

Lecture 13
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Outline – Implicit and Diffusion Monte Carlo Techniques

Why explicit MCRT becomes difficult

Two failure modes Chapter 10 addresses:

1. **Optically thick transport.** In a scattering region of size L and mean free path ℓ, packet motion is a random walk:

   $$\tau = L/\ell, \qquad N_{\text{step}} \sim \left(\frac{L}{\ell}\right)^2 = \tau^2.$$

   A cell with $\tau = 100$ needs $\sim 10^4$ interactions per packet just to diffuse across.
2. **Thermal radiative transfer (TRT).** Radiation and matter exchange energy on the absorption-reemission timescale. An explicit time step that lets radiation drain matter faster than the material temperature is updated can produce negative internal energies.

The question. What physically justified shortcut can replace many microscopic events without losing the essential radiative-transfer behavior?

Two families of techniques

- **Implicit Monte Carlo (IMC).**
  Replace most rapid absorption-reemission cycles by *effective scattering*, leaving only the net imbalance as true matter-radiation energy exchange.
  $\Rightarrow$ addresses the *time-step* problem.

- **Diffusion-based acceleration.**
  In optically thick regions, replace the explicit packet history by a probabilistic diffusion step:
  - Modified Random Walk (MRW) – a diffusion sphere inside one cell.
  - Discrete Diffusion Monte Carlo (DDMC) – a whole cell treated as a diffusion zone.

  $\Rightarrow$ addresses the *spatial* cost of thick transport.

A one-dimensional gray TRT model

Following Fleck & Cummings (1971): let $I(x,t;\mu)$ be the specific intensity, χ the absorption opacity, U the material internal energy density, and

$$E = a_{\mathrm{R}} T^4, \qquad \beta = \frac{\partial E}{\partial U}.$$

Then

$$\frac{1}{c}\frac{\partial I}{\partial t} + \mu \frac{\partial I}{\partial x} + \chi I = \frac{1}{2}\chi c E,$$

$$\frac{1}{\beta}\frac{\partial E}{\partial t} = \chi \int_{-1}^{1} I \,\mathrm{d}\mu \; - \; \chi c E \; + \; S.$$

Radiation is transported, absorbed, and thermally reemitted; the factor $1/2$ is the isotropic emissivity in 1D (4π becomes the angular length 2).

What goes wrong, and the IMC fix

Explicit treatment. The emission $\propto cE^n = ca_{\mathrm{R}}(T^n)^4$ is held fixed during the step. When χ, β, or Δt is large, this is a bad approximation: matter and radiation actually exchange energy many times within one step.

The IMC reinterpretation

Treat most absorption-reemission cycles as *effective scatterings*; keep only the net imbalance as true absorption.

Physical picture.

$$\text{packet absorption} \;\rightarrow\; \text{matter energy gain} \;\rightarrow\; \text{thermal reemission.}$$

If absorption and reemission happen within one Δt, the cycle represents no lasting matter energy change. IMC groups it as a single scattering-like event. The opacity is split:

$$\chi_{\mathrm{eff}}^{\mathrm{abs}} = f\chi, \qquad \chi_{\mathrm{eff}}^{\mathrm{scat}} = (1-f)\chi.$$

Deriving the Fleck factor

Time-center the equilibrium energy: $\bar{E} = \alpha E^{n+1} + (1-\alpha)E^n$, $0 \leq \alpha \leq 1$. Substitute into the discretized material equation

$$\frac{E^{n+1} - E^n}{\beta \Delta t} = \chi \bar{\mathscr{I}} - \chi c \bar{E} + S, \qquad \bar{\mathscr{I}} \equiv \int_{-1}^{1} \bar{I}\, \mathrm{d}\mu.$$

Collecting E^{n+1} on the left,

$$(1 + \alpha\beta c \Delta t \chi) E^{n+1} = E^n + \beta \Delta t \left[\chi \bar{\mathscr{I}} + S - (1-\alpha)\chi c E^n \right],$$

and the inverse coefficient is the Fleck factor

$$\boxed{f = \frac{1}{1 + \alpha\beta c \Delta t \chi}}.$$

Substituting $\bar{E}$ back into the transport equation gives a source split (using $1 - f = \alpha\beta c \Delta t \chi f$):

$$\tfrac{1}{2}\chi c \bar{E} = \tfrac{1}{2} f \chi c E^n + \underbrace{\tfrac{1}{2}(1-f)\chi \bar{\mathscr{I}}}_{\text{isotropic scattering}} + \tfrac{1}{2}\alpha f \beta \chi c \Delta t S.$$

Reading the Fleck factor

$$f = \frac{1}{1 + \alpha\beta c\Delta t\chi}$$

- $\alpha\beta c\Delta t\chi \ll 1 \;\Rightarrow f \simeq 1$: the scheme reduces to explicit MCRT.
- Large Δt, large χ, or strong coupling $\beta \Rightarrow f \ll 1$: most interactions are *effective scatterings*.
- $\alpha \geq 1/2$ gives unconditional time-step stability.

Cost. Stability is not the same as accuracy. Time-discretization error remains; in optically thick regions $f \ll 1$ means many effective scatterings, so IMC alone does not solve the spatial cost.

Effective opacities as a function of $\beta c \Delta t \chi$

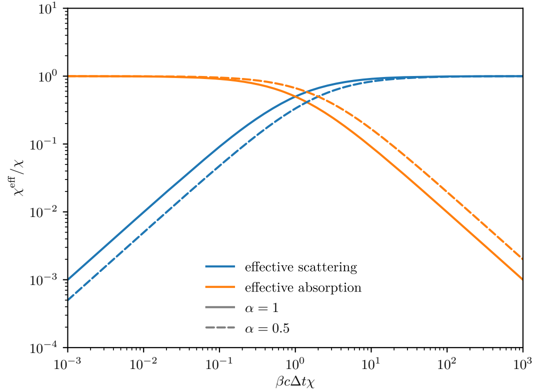

Figure 1: Effective scattering $(1-f)\chi$ and effective absorption $f\chi$, normalized by the true opacity χ, versus $\beta c \Delta t \chi$ for two choices of α. At large $\beta c \Delta t \chi$ the effective scattering opacity is orders of magnitude larger than the residual net absorption. Reproduced from Noebauer & Sim (2019), Fig. 14.

Known weaknesses of IMC

- **Maximum-principle violation.**
  With no internal source, interior temperatures can exceed the imposed boundary temperatures (Larsen & Mercer 1987). Their stability-restoring Δt bound can be quite tight in practice.
- **Teleportation error.**
  When $f \ll 1$, much of the opacity acts as effective scattering, and packets can move farther than the true diffusion limit would allow. A numerical artifact of the effective-scattering reinterpretation.
- **Other IMC variants.**
  Symbolic Implicit Monte Carlo (SIMC, Brooks 1989) keeps the thermal emission term formally unknown via symbolic packet weights – closer to a truly implicit method, but more complex.

Random walk to diffusion

Mean free path ℓ, exponential free-path distribution:

$$\langle s \rangle = \ell, \quad \langle s^2 \rangle = 2\ell^2.$$

After N isotropically directed steps, cross terms average to zero:

$$\langle r^2 \rangle = 2N\ell^2, \qquad t \simeq N\ell/c \implies \langle r^2 \rangle = 2c\ell\, t.$$

Matching the 3D diffusion form $\langle r^2 \rangle = 6D_\mathrm{t} t$,

$$\boxed{D_\mathrm{t} = \frac{c\ell}{3} = \frac{c}{3\chi_\mathrm{R}}}.$$

Escape from a region of size L needs $N \sim (L/\ell)^2 = \tau^2$ steps – the same scaling we saw as the motivation for diffusion acceleration.

Moment-equation derivation of D_t

Take zeroth and first angular moments of the RT equation; close the second moment with the Eddington approximation $P_{ij} = \frac{1}{3} E \delta_{ij}$:

$$\frac{1}{c}\frac{\partial E}{\partial t} + \nabla\cdot\mathbf{F} = 0, \qquad \frac{1}{c}\frac{\partial \mathbf{F}}{\partial t} + \frac{c}{3}\nabla E = -\chi_\mathrm{R}\,\mathbf{F}.$$

Drop the flux time derivative (diffusion-limit assumption):

$$\mathbf{F} = -\frac{c}{3\chi_\mathrm{R}}\nabla E = -D_\mathrm{t}\nabla E, \qquad \frac{\partial E}{\partial t} = \nabla\cdot(D_\mathrm{t}\nabla E).$$

Where the $1/3$ comes from. It is the Eddington factor $\langle\mu^2\rangle = \int_{-1}^{1}\mu^2\,\mathrm{d}\mu/2 = 1/3$ of an isotropic radiation field – the same $d = 3$ of the random-walk formula $\langle r^2\rangle = 2dD_\mathrm{t}t$. Both express the loss of directional memory in the deep interior.

MRW: the basic procedure

Origin: Fleck & Canfield (1984); adapted for dust RT by Min et al. (2009) and Robitaille (2010).

1. Around the current packet position, build the largest *diffusion sphere* that fits entirely inside the current grid cell. Radius R_0; Rosseland mean free path $\ell_{\mathrm{R}} = 1/\chi_{\mathrm{R}}$.
2. Activate the diffusion shortcut if the sphere is optically thick enough *and* the packet would actually scatter inside it:
$$R_0 > \ell_{\mathrm{R}}, \qquad \ell_{\mathrm{col}} < R_0.$$
3. Sample one stochastic diffusion outcome: an escape time, an exit point on the sphere surface, and a new direction. The many small interactions inside R_0 are compressed into one move.

The MRW geometry

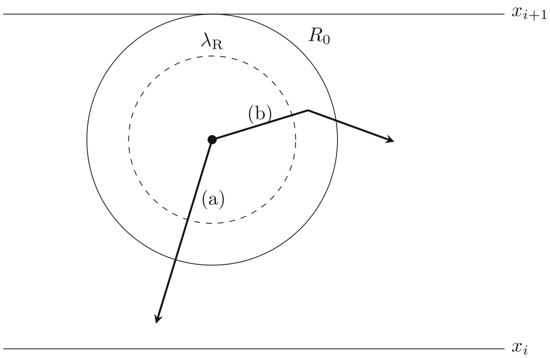

Figure 2: Construction of the diffusion sphere. The thick dot is the packet position; the solid circle (radius R_0) is the largest cell-enclosed sphere; the dashed circle has radius λ_R. (a) The sampled free path ℓ_{col} leaves the sphere without interaction: use ordinary transport. (b) Both activation criteria are met: switch on the diffusion shortcut. Reproduced from Noebauer & Sim (2019), Fig. 15, after Fleck & Canfield (1984).

Survival probability of the absorbing sphere

Diffusion equation with an absorbing boundary:

$$\partial_t p = D_\mathrm{t}\nabla^2 p, \qquad p(R_0,t)=0, \qquad p(\mathbf{r},0)=\delta^3(\mathbf{r}).$$

Substitute $u(r,t)=r\,p(r,t)$: a 1D heat equation on $[0,R_0]$ with Dirichlet ends. Eigenfunctions $\sin(n\pi r/R_0)$, decay rates $n^2\pi^2 D_\mathrm{t}/R_0^2$. Radial density:

$$\psi(r,t)=\frac{1}{R_0^2}\sum_{n=1}^{\infty}\frac{n}{r}\exp\left(-\frac{n^2\pi^2 D_\mathrm{t} t}{R_0^2}\right)\sin\left(\frac{n\pi r}{R_0}\right).$$

Survival probability (Fleck & Canfield 1984; Carslaw & Jaeger §9.3):

$$P_\mathrm{R}(t)=4\pi\int_0^{R_0}\psi(r,t)\,r^2\,\mathrm{d}r = 2\sum_{n=1}^{\infty}(-1)^{n+1}\exp\left(-\frac{n^2\pi^2 D_\mathrm{t} t}{R_0^2}\right).$$

Sampling the escape time

Draw $\xi \in (0,1)$. If $\xi \le P_T(t) \equiv 1 - P_{\mathrm{R}}(t)$, the packet has *escaped*; reuse the same ξ to set the escape time:

$$P_T(t_{\mathrm{esc}}) = \xi \quad \text{(inverted by bisection or a precomputed table).}$$

Otherwise the packet stays inside; its new radius is drawn from the surviving distribution by inverting

$$4\pi \int_0^{R_1} \psi(r,t)\, r^2\, \mathrm{d}r = \xi\, P_{\mathrm{R}}(t).$$

Mean escape time. Solving $D_{\mathrm{t}} \nabla^2 u = -1$ with $u(R_0) = 0$ gives $u(r) = (R_0^2 - r^2)/(6D_{\mathrm{t}})$, so

$$\langle t_{\mathrm{esc}} \rangle = \int_0^\infty P_{\mathrm{R}}(t)\, \mathrm{d}t = \frac{2R_0^2}{\pi^2 D_{\mathrm{t}}} \sum_{n\ge 1} \frac{(-1)^{n+1}}{n^2} = \frac{R_0^2}{6D_{\mathrm{t}}}.$$

The series for $P_{\mathrm{R}}(t)$ converges geometrically once $\pi^2 D_{\mathrm{t}} t / R_0^2 \gtrsim 1$; two or three terms suffice.

Algorithm: MRW step for dust RT (1/2)

```
def mrw_step(packet, cell, dust):

    # (0) Ordinary free-path sample; needed for activation test.
    tau     = -log(uniform(0, 1))
    chi_nu  = dust.chi(packet.nu, cell.T_dust)        # total ext. at nu
    ell_col = tau / chi_nu

    # (1) Largest cell-enclosed sphere around the packet.
    R_0   = distance_to_nearest_cell_face(packet.r, cell)
    chi_R = dust.chi_rosseland(cell.T_dust)
    ell_R = 1.0 / chi_R
    D_t   = c / (3.0 * chi_R)

    # (2) Activation: R_0 > ell_R AND ell_col < R_0.
    if not (R_0 > ell_R and ell_col < R_0):
        return propagate_ordinary(packet, ell_col)

    # (3) Sample escape time from absorbing-sphere diffusion.
    #     P_R(t) = 2 * sum_{n>=1} (-1)^(n+1)
    #                  * exp(-n^2 pi^2 D_t t / R_0^2).
    #     Invert xi = 1 - P_R(t_esc) (table or bisection).
    xi    = uniform(0, 1)
    t_esc = invert_PT(xi, R_0, D_t)
    # Sanity: <t_esc> = R_0^2 / (6 D_t).
```

Algorithm: MRW step for dust RT (2/2)

```
# (4) Dust-heating estimator: effective path length c * t_esc
#     Lucy / Bjorkman-Wood path-length estimator with the
#     Planck-mean absorption opacity.
kappa_abs_P      = dust.kappa_abs_planck(cell.T_dust)
cell.E_absorbed += packet.L * kappa_abs_P * c * t_esc

# (5) Exit position: uniform on the sphere surface.
n_out    = sample_unit_vector_uniform_on_sphere()
packet.r = packet.r + R_0 * n_out

# (6) Exit direction: Lambertian about the outward normal,
#     p(mu) d mu = 2 mu d mu, with mu = k . n_out in [0, 1].
mu       = sqrt(uniform(0, 1))
phi      = 2 * pi * uniform(0, 1)
packet.k = rotate_into_frame(n_out, mu, phi)

# (7) New frequency from the local dust thermal emissivity:
#     p(nu) ~ kappa_abs(nu, T_dust) * B_nu(T_dust).
packet.nu = sample_from_dust_emissivity(dust, cell.T_dust)
```

Note. D_t uses the Rosseland mean (frequency-integrated diffusion). Heating uses the Planck-mean absorption opacity. The two can differ by an order of magnitude in optically thick dust.

MRW in action

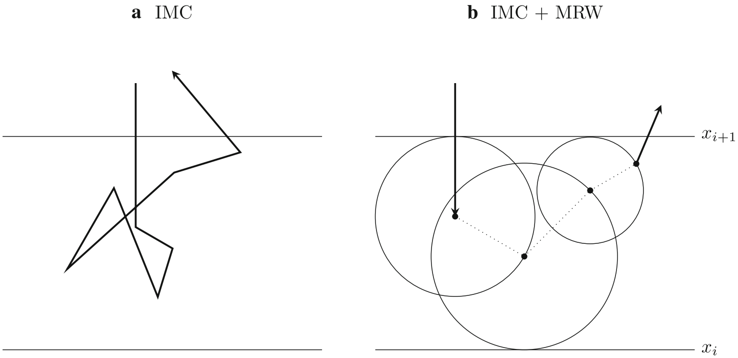

Figure 3: (a) IMC alone: every effective scattering is followed explicitly. (b) IMC + MRW: each time the activation criteria are met, the per-scattering transport (thick solid) is replaced by one diffusion step across a sphere (thin dotted) terminated by a randomly sampled exit point on the sphere surface (black dot). Near the cell faces x_i, x_{i+1} the largest fitting sphere is too small and ordinary transport resumes. Reproduced from Noebauer & Sim (2019), Fig. 16.

Note on resonance-line transfer

Why MRW is *not* the standard accelerator for resonance lines (e.g. Lyα):

- Trapping is *frequency*-driven: the opacity has a Voigt profile that varies sharply with ν.
- A fixed-radius diffusion sphere with a single D_{t} is no longer a good approximation.

The dominant acceleration techniques in resonance-line MCRT instead act in frequency space:

- **Core-skipping** (Ahn, Lee & Lee 2002): force the packet into the line wings, where the optical depth drops steeply.
- **Analytic Gaussian frequency steps** (Loeb & Rybicki 1999): a Fokker–Planck-like treatment of resonant scattering in expanding media.

Strengths and limitations of MRW

Strengths.

- Local. Adds to an existing transport code without changing the cell-to-cell event loop.
- Slots cleanly into the Lucy / Bjorkman-Wood path-length temperature estimator (step 4 of the algorithm).
- Standard in every major dust RT code: HYPERION, SKIRT, RADMC-3D, MCFOST, TORUS.

Limitations.

- Near a cell boundary the largest fitting sphere is small; activation fails and ordinary MC takes over.
- Complicated frequency-dependent or Sobolev-type opacities break the single-D_{t} picture.
- Per-packet exit direction is approximated as Lambertian; the true Green's-function exit bias is small and ignored.

From MRW to DDMC

MRW. A spherical diffusion subregion, sized to fit inside one grid cell. Near cell faces the sphere becomes small.

DDMC. Treat an entire optically thick grid cell as a diffusion region (Densmore et al. 2007). The packet no longer carries a precise position and direction; its state variable is the host cell. It undergoes stochastic *leakage* from cell to cell, or true absorption.

Related. Implicit Monte Carlo Diffusion (IMD) of Gentile (2001) discretizes the time derivative of the diffusion equation by finite differences *in addition* to the spatial discretization done by DDMC. Multifrequency / multigroup descendants exist in both flavours (Densmore et al. 2012; Wollaeger et al. 2013).

The discretized diffusion equation

Gray 1D, finite-volume:

$$\frac{1}{c}\frac{\partial \phi}{\partial t} = \frac{\partial}{\partial x}\left(D\frac{\partial \phi}{\partial x}\right) - f\sigma\phi + q, \qquad D = \frac{1}{3\sigma_{\mathrm{R}}}.$$

Interface flux: $F_{j+1/2} = -D_{j+1/2}\,(\phi_{j+1} - \phi_j)/\delta x_{j+1/2}$. Integrating over cell j (width Δx_j) gives the DDMC equation

$$\frac{1}{c}\frac{\mathrm{d}\phi_j}{\mathrm{d}t} + (\sigma_{L,j} + \sigma_{R,j} + f_j\sigma_j)\,\phi_j = q_j + \text{leakage from neighbors},$$

with leakage opacities

$$\sigma_{R,j} = \frac{D_{j+1/2}}{\Delta x_j \delta x_{j+1/2}}, \qquad \sigma_{L,j} = \frac{D_{j-1/2}}{\Delta x_j \delta x_{j-1/2}}.$$

Sampled like an ordinary collision:

$$t_{\mathrm{col}} = \frac{-\ln\xi}{c\,(\sigma_{L,j} + \sigma_{R,j} + f_j\sigma_j)},$$

with event type chosen in proportion to the three opacity-like terms.

Hybrid transport / diffusion

Realistic problems mix thin and thick regions, so DDMC is used in a hybrid with IMC or ordinary transport.

- **Transport $\to$ DDMC interface.** A boundary-condition-based probability decides whether the incoming packet becomes a DDMC particle or is reflected. Densmore et al. (2007) used the asymptotic diffusion-limit boundary condition, which is more accurate than a Marshak boundary when the incoming distribution is anisotropic.
- **DDMC $\to$ transport interface.** A leaked DDMC particle is placed at the cell face and converted back to a transport packet with a sampled direction.

Domain. DDMC is the standard accelerator in supernova / thermal RT codes (SuperNu, JAYENNE / Capsaicin), where multifrequency / multigroup machinery is essential.

Why DDMC is rare in dust RT

Every major dust RT code uses MRW, not DDMC. Three practical reasons:

1. **Invasiveness.**
   MRW needs only the local cell-enclosed sphere and one Rosseland-mean opacity. DDMC discretizes diffusion on the mesh: leakage opacities on every face, transport-diffusion interfaces, modified packet bookkeeping.
2. **Frequency dependence.**
   Dust opacity collapses to Planck / Rosseland means – MRW uses the latter directly. Multifrequency DDMC needs per-group leakage opacities and cross-group coupling: a significant implementation cost.
3. **Estimator compatibility.**
   Dust RT temperatures rely on the Lucy / Bjorkman-Wood path-length estimator. MRW provides a path length $c\,t_{\rm esc}$ naturally (step 4). DDMC's cell-discrete leakage formulation has no path-length analogue and needs separate deposition rules.

Domain choice is driven by code architecture, not by which method is theoretically more powerful.

Side-by-side comparison

Method	Problem addressed	Core idea	Main caveats
IMC	Time instability in TRT	Keep only $f\chi$ as true absorption; treat $(1-f)\chi$ as effective scattering	Stable for large Δt, but suffers from time-discretization error, maximum-principle violations, and teleportation
MRW	Many scattering steps inside one optically thick cell	Build a cell-enclosed diffusion sphere; replace the internal random walk by one diffusion step	Inefficient near cell boundaries; difficult with strongly varying opacities
DDMC	Diffusive transport across optically thick regions	Treat whole cells as diffusion zones; sample cell-to-cell leakage from a discretized diffusion equation	Requires careful interface conditions and diffusion-region selection; more invasive than MRW

These methods are not alternatives: IMC targets *time*, MRW and DDMC target *space*. In production codes they are typically combined.

Take-home messages

Fleck factor.

$$f = \frac{1}{1 + \alpha\beta c\Delta t \chi}$$

fixes the IMC time-step problem by reinterpreting most rapid cycles as effective scatterings.

Diffusion shortcut. In the optically thick limit the exact sequence of microscopic scatterings is less important than the probability of escape through a boundary. MRW and DDMC exploit this fact.

No free lunch. Each acceleration has a regime of validity:

- IMC: large Δt, at the cost of time-discretization error.
- MRW: deep inside thick cells, weak near boundaries.
- DDMC: thick regions, but interface handling is essential.

Always check time-step convergence and benchmark against an unaccelerated reference.

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- Densmore et al. 2007, *JCP* **222**, 485; Densmore et al. 2012, *JCP* **231**, 6924 – DDMC.
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- Wollaber 2016, *JCTT* – four-decade review of IMC.
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